AYBERK YARKIN YILDIZ

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Summary

Researcher focusing on model-distributed inference via communication-aware pruning, mapping, and compression, and Markovian experimental design for robust re-training under concept drift. Experienced in building ML systems across sensing modalities, including wireless radar, RF signal sensing, sEMG motion analysis, and multivariate time-series modeling.

Education

Northeastern University — *Boston, MA, USA*
Ph.D. in Computer Engineering — *Expected April 2028*
M.Sc. in Electrical and Computer Engineering — *April 2025*

Bilkent University — *Ankara, Turkey*
B.Sc. in Electrical and Electronics Engineering — *July 2023*

Friedrich-Alexander-Universität — *Erlangen, Germany*
Erasmus Student in Elektrotechnik – Elektronik und Informationstechnik — *August 2022*

Mehmet Emin Resulzade — *Ankara, Turkey*
High School Degree — *June 2018*

Skills

Programming Languages: Python, MATLAB, C++
Frameworks: PyTorch, Tensorflow, PySpark, Slurm, Bash
Languages: English (Fluent – C1 level), Turkish (Native Fluency), German (Beginner – A2 level)
Selected Courses: Advanced Machine Learning, Verifiable Machine Learning, Probabilistic System Modeling, Reinforcement Learning and Decision Making Under Uncertainty, Advanced Computer Vision, Machine Learning with Small Data

Experience

Northeastern University — *Boston, MA, USA*
Graduate Research Assistant — *Sep 2023 – April 2028*
Research Labs: DNAL, WIoT, SPIRAL. **Advisor:** Prof. Stratis Ioannidis

- Designed Markovian experimental design algorithms for optimal data selection to address performance degradation of ML models under concept drift, resulting in improved predictive accuracy with provable optimality guarantees.
- Developed scalable, communication-aware optimization methods for neural network pruning and mapping to mitigate bandwidth, latency, and energy constraints in distributed systems, enabling resilient and low-latency inference across heterogeneous hardware topologies.

Neurocess Limited — *London, UK*
Data Science / Machine Learning Engineer (Remote) — *Sep 2022 – July 2023*

- Built Siamese CNN–Transformer architectures for multivariate sEMG time-series analysis to solve athlete performance monitoring under limited data, achieving similarity-based classification and anomaly detection in PyTorch and Keras.

KOCLAB, National Magnetic Resonance Research Center (UMRAM) — *Ankara, Turkey*
Undergraduate Researcher — *Sep 2021 – Sep 2022*

- Implemented deep learning models based on RNNs and Transformers to address missing data and temporal dependencies in biomedical time-series, producing imputation and forecasting pipelines validated in PyTorch experiments.

TUBITAK SAGE — *Ankara, Turkey*
Intern — *June 2021 – July 2021*

- Designed nano-drone hardware and software simulations in Altium Designer to improve agility and reduce detectability in defense applications, supporting early-stage validation of stealth-oriented aerial platforms.

UMRAM — *Ankara, Turkey*
Intern — *July 2020 – Aug 2020*

- Implemented and tested MATLAB-based interfaces for fundamental electronic devices (gaussmeter, analog filter, and DC power supply) to support scalable experimentation for future research, resulting in a publication in *Medical Physics*.

Projects

Markovian Experimental Design under Concept Drift — *2024 – Present*

- Implemented Kalman-filter-based data selection strategies to combat concept drift in sequential learning settings, demonstrating sustained predictive performance in ML models under non-stationary data distributions.

Communication-aware neural Mapping and Pruning Framework *2024 – Present*

- Developed a communication-aware pruning and model-mapping pipeline to reduce inference latency and network overhead in distributed CNNs, achieving up to 26× speedup on real-world testbeds including Colosseum and Raspberry Pi clusters.

Multi-Agent Distributed Inference over Large AI Models at the Network Edge *2025*

- Designed a compression-driven Model-Distributed Inference (MDI) framework to enable execution of large models across heterogeneous edge devices, reducing wireless communication overhead via lightweight pruning and quantization.

Gradient Boosting Decision Trees on Medical Diagnosis View publication [4] *2024*

- Conducted large-scale empirical analysis of ensemble models to evaluate performance trade-offs against deep learning methods on tabular medical data, showing consistent superiority of GBDTs across multiple benchmarks.

Wireless Radar Classification with Transformers View preprint [5] *2023 – 2024*

- Implemented Transformer-based radar signal classifiers with LoRA and conformal prediction to improve robustness under distribution shifts, achieving reliable OOD performance in PyTorch.

Portable RF Signal Sensing System Using SDR View publication [6] *2023*

- Implemented a GPU-accelerated SDR-based Electronic Support Measures (ESM) system to address real-time RF signal awareness requirements, enabling efficient detection, measurement, and classification using GNU Radio and XGBoost.

sEMG Motion Classification and Anomaly Detection View preprint *2022 – 2023*

- Developed few-shot Siamese learning models to address limited-label sEMG motion classification, achieving robust generalization across unseen motion patterns.

Multivariate Time Series Imputation With Transformers View publication [7] *2022*

- Designed a Transformer-based autoencoder to solve missing-value imputation in multivariate time series, outperforming seven state-of-the-art methods by 13.5–50.5% over benchmark datasets.

Publications

[1] **A. Y. Yıldız**, L. Su, C. Joe-Wong, E. Yeh, and S. Ioannidis, "Markovian Experimental Design under Concept Drift." (submitted to ICML 2026)

[2] **A. Y. Yıldız***, M. Sirera Perelló*, T. T. Davarakis*, J. Groen, H. Seferoglu, K. Chowdury, and S. Ioannidis, "CaMP: Communication-Aware Neural Mapping and Pruning." (submitted to ACM SIGMETRICS 2026)

[3] **A. Y. Yıldız***, M. Sirera Perelló*, T. Li*, C. Li, P. Gholami, S. Reddy, T. T. Davarakis, A. Arora, K. Chowdhury, A. Eryılmaz, S. Ioannidis, H. Seferoglu, and N. Shroff, "Multi-Agent Distributed Inference over Large AI Models at the Network Edge." (submitted to IEEE Wireless Communications Magazine)

[4] **A. Y. Yıldız** and A. Kalayci, "Gradient boosting decision trees on medical diagnosis over tabular data." 2025 IEEE International Conference on AI and Data Analytics (ICAD). IEEE, 2025.

[5] M. Belgiovine, J. Groen, M. Sirera, C. Tassie, **A. Y. Yıldız**, S. Ioannidis, and K. Chowdhury, "T-PRIME: Real-time Deployment of a Transformer-based Protocol Identification for Machine-learning at the Edge." arXiv preprint arXiv:2401.04837 (2024). (submitted to IEEE/ACM Transactions on Networking)

[6] G. S. Yavuz, B. Sayğılı, Y. Aydınlı, R. Dalkıran, İ. Eşin, M. Uluçay, B. Uykulu, S. S. Kıyma, O. Arikan, and **A. Y. Yıldız**, "Detection and classification architecture for sdr based radar electronic support measure systems." 2024 32nd Signal Processing and Communications Applications Conference (SIU). IEEE, 2024.

[7] **A. Y. Yıldız**, E. Koç, and A. Koç. "Multivariate time series imputation with transformers." IEEE Signal Processing Letters 29 (2022): 2517-2521.

Honors & Awards

- Invited Talks: IEEE SPS, IEEE ICIP *2023 – 2024*
- IEEE SPS's top 25 downloaded articles *2022 – 2023*
- Research Excellence Award at Bilkent University *2023*
- High Honor Student & Tuition Scholarship at Bilkent University *2018 – 2023*
- 8/8th grade practical and theoretical certificate for piano, ABRSM, London, UK *2023*

Services

- Technical Reviewer: International Conference on Artificial Intelligence and Statistics, IEEE International Conference on AI and Data Analytics
- Amazon IET-MSI Program Participant *Oct 2024*
- Northeastern University SPIRAL Committee Member *Dec 2024*
  - Organized seminars with participants from universities in USA and Switzerland.